



THE UNIVERSITY OF
MELBOURNE

OpenStack and Linux



APT (Advanced Packaging Tool)

- Fetch package list, update package index
 - *sudo apt update*
- Update packages
 - *sudo apt upgrade*
- Install packages
 - *sudo apt install <some package>*
- apt vs apt-get
 - *apt*: simpler, modern, and more intuitive
 - *apt-get*: script friendly, backward compatibility, and more predictable outputs



Package managers - macOS

Homebrew (works for Linux too!)

- Install packages
 - *brew install <some package>* <- And the dependencies
- Install macOS native applications
 - *brew install --cask temurin* <- Install Temurin JDK
- Update packages
 - *brew upgrade*
- Fetch package list, update package index
 - *brew update* <- Now run before brew install automatically!
- Clean up
 - *brew cleanup* <- Now run after brew install automatically!





Package managers - Homebrew



Brew on Linux (and WSL²)

- Docs:
 - [Homebrew on Linux](#)
 - [Alternative installation](#)
- TL;DR
 - Use "Alternative installation" on password-less Linux, then
 - `(echo; echo 'eval "$(homebrew/bin/brew shellenv)') >> ~/.bashrc`
 - `eval "$(homebrew/bin/brew shellenv)"`
 - For WSL2
 - `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
 - `(echo; echo 'eval "$(brew shellenv)"' >> ~/.bashrc`
 - `eval "$(brew shellenv)"`



Get ready for assignment 2 - jq

jq – a lightweight and flexible JSON processor

- Installation

- *brew install jq* (HomeBrew)
- *sudo apt install jq* (Ubuntu)

- Examples

- *jq '.employees[].name' data.json*
- *jq '.employees += [{"name": "Jane Doe", "age": 25, "department": "Design"}]' data.json*
- *jq '.employees += [{"name": "Jane Doe", "age": 25, "department": "Design"}]' data.json > tmp.json && mv tmp.json data.json*
- *jq '.employees |= map(select(.name != "Jane Doe"))' data.json*

```
{  
  "employees": [  
    {  
      "name": "John Doe",  
      "age": 30,  
      "department": "Engineering"  
    }  
  ]  
}
```



Get ready for assignment 2 – asdf

asdf – manage all runtime versions with one tool

- Install asdf
 - HomeBrew
 - *brew install asdf*
 - *echo -e "\n. \$(brew --prefix asdf)/libexec/asdf.sh" >> ~/.profile*
 - *echo -e "\n. \$(brew --prefix asdf)/etc/bash_completion.d/asdf.bash" >> ~/.profile*
 - Ubuntu
 - *sudo apt install curl git*
 - *git clone https://github.com/asdf-vm/asdf.git ~/.asdf --branch v0.14.0*
 - *(echo; echo '. "\$HOME/.asdf/asdf.sh") >> ~/.bashrc*
 - *(echo; echo '. "\$HOME/.asdf/completions/asdf.bash") >> ~/.bashrc*



Get ready for assignment 2 – asdf

- Install [asdf plugins](#)
 - *asdf plugin add kubectl*
 - *asdf plugin add helm*
 - *asdf plugin add fission*
- [asdf cheat sheet](#)
- kubectl (1.33.8)
 - Install kubectl version 1.33.8
 - *asdf install kubectl 1.33.8*
 - Set the local version
 - *asdf local kubectl 1.33.8*



Get ready for assignment 2 – asdf

- Helm (3.18.6)
 - Install helm version 3.18.6
 - *asdf install helm 3.18.6*
 - Set the local version
 - *asdf local helm 3.18.6*
- Fission (1.20.0)
 - Install Fission version 1.20.0
 - *asdf install fission 1.20.0*
 - Set the local version
 - *asdf local fission 1.20.0*



Get ready for assignment 2 – Pipeline

- Unix Pipeline - output of one command is passed as input to the next
 - *openstack flavor list*

```
> openstack flavor list
```

ID	Name	RAM	Disk	Ephemeral	VCPUs	Is Public
14537b92-df17-4cb8-850c-7750c7f59d29	p3.medium	8192	30	0	4	True
2dc4e594-bad9-49ff-ae19-ef9391b4e2fb	p3.xsmall	2048	30	0	1	True
3925a224-11e9-4f8b-9324-04d7fd3bc44e	p3.large	16384	30	0	8	True
406352b0-2413-4ea6-b219-1a4218fd7d3b	t3.xsmall	1024	10	0	1	True
43704bde-9b6e-459f-81d7-9a483b875c13	p3.xlarge	32768	30	0	16	True
501142ac-ab11-4854-96c3-03050074851a	m3.large	16384	30	0	8	True
5d8b8337-dc22-4ac7-9d4c-fda749d364bf	uom.mse.2c9g	9216	30	0	2	False
71359ce0-49c8-4485-94ac-7936703c961b	t3.small	2048	10	0	2	True
78d89b0a-f126-4e13-ab09-a2df697cc61a	uom.mse.16c64g	65536	30	0	16	False
7e745eed-bcc3-44b3-ab22-10275f0e4759	uom.mse.1c4g	4608	30	0	1	False
8b3bac8d-a44f-40f2-8516-533c18030434	m3.xlarge	32768	30	0	16	True
92e5a7b0-25a2-43bb-83c3-f714e7b58203	m3.small	4096	30	0	2	True
943f19c5-2c29-4beb-96da-3ff7379c2edd	p3.xxlarge	65536	30	0	32	True
951fb7e9-6517-43fa-bf2b-a03e9604e936	t3.medium	4096	10	0	4	True
a23c0455-c032-445d-ba19-94e92771edf9	uom.mse.12c48g	49152	30	0	12	False
b469c55d-7266-4172-abfb-9859e7a79990	uom.mse.32c128g	131072	30	0	32	False
b7336043-bcba-4e5b-bdae-b76ccc328168	m3.medium	8192	30	0	4	True
cc7ab7b9-040c-46ec-bac4-7d31a3633c75	p3.small	4096	30	0	2	True
ebd9c273-58f3-48f4-8e0d-e4435169c3ab	uom.mse.24c96g	98304	30	0	24	False
eeff3572-69d0-4a72-b176-b77f2fe1ba19	m3.xsmall	2048	30	0	1	True
f08d61e5-edb2-4a0d-ba14-d1526741d346	m3.xxlarge	65536	30	0	32	True



Get ready for assignment 2 – Pipeline

- Unix Pipeline - output of one command is passed as input to the next

– *openstack flavor list | grep 'uom.mse.2c9g'*

```
> openstack flavor list | grep 'uom.mse.2c9g'
| 5d8b8337-dc22-4ac7-9d4c-fda749d364bf | uom.mse.2c9g | 9216 | 30 | 0 | 2 | False |
```

– *openstack flavor list | grep 'uom.mse.2c9g' | awk '{print \$2}'*

```
> openstack flavor list | grep 'uom.mse.2c9g' | awk '{print $2}'
5d8b8337-dc22-4ac7-9d4c-fda749d364bf
```

– *openstack flavor list | grep 'uom.mse.2c9g' | awk '{print \$1}'*

– *openstack flavor list | grep 'uom.mse.2c9g' | awk '{print \$4}'*



Get ready for assignment 2 – Command Substitution

- Command Substitution – executes the command inside the brackets and then replaces the entire '\$(...)' expression with the STDOUT of the command

```
openstack coe cluster template create \  
  --keypair mykeypair \  
  --labels "container_infra_prefix=registry.rc.nectar.org.au/nectarmagnum/; \  
master_lb_floating_ip_enabled=false;\  
cinder_csi_enabled=true;\  

```

```
--master-lb-enabled \  
--master-flavor=$(openstack flavor list | grep 'uom.mse.2c9g' | awk '{print $2}') \  
--flavor=$(openstack flavor list | grep 'uom.mse.2c9g' | awk '{print $2}') \  
--server-type=vm \  

```

– *--master-flavor=5d8b8337-dc22-4ac7-9d4c-fda749d364bf \
--flavor=5d8b8337-dc22-4ac7-9d4c-fda749d364bf *



Get ready for assignment 2 – Command Output

- OpenStack command outputs

- *openstack flavor list*

```
> openstack flavor list
```

ID	Name	RAM	Disk	Ephemeral	VCPUs	Is Public
14537b92-df17-4cb8-850c-7750c7f59d29	p3.medium	8192	30	0	4	True
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3925a224-11e9-4f8b-9324-04d7fd3bc44e	p3.large	16384	30	0	8	True
406352b0-2413-4ea6-b219-1a4218fd7d3b	t3.xsmall	1024	10	0	1	True
43704bde-9b6e-459f-81d7-9a483b875c13	p3.xlarge	32768	30	0	16	True
501142ac-ab11-4854-96c3-03050074851a	m3.large	16384	30	0	8	True

- *openstack flavor list -f value -c ID -c Name*

```
> openstack flavor list -f value -c ID -c Name
14537b92-df17-4cb8-850c-7750c7f59d29 p3.medium
2dc4e594-bad9-49ff-ae19-ef9391b4e2fb p3.xsmall
3925a224-11e9-4f8b-9324-04d7fd3bc44e p3.large
406352b0-2413-4ea6-b219-1a4218fd7d3b t3.xsmall
43704bde-9b6e-459f-81d7-9a483b875c13 p3.xlarge
501142ac-ab11-4854-96c3-03050074851a m3.large
```

- *openstack flavor list -f value -c ID -c Name | grep 'uom.mse.2c9g' | awk '{print \$1}'*
 - *openstack flavor list -f json -c ID -c Name | jq -r '[]' | select(.Name == "uom.mse.2c9g") | .ID'*



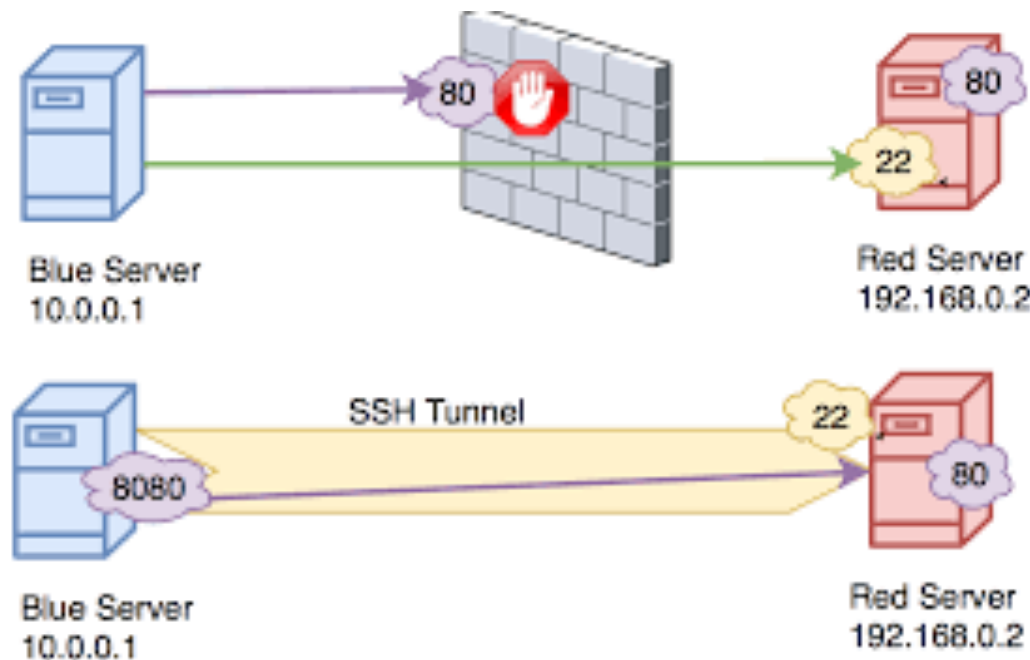
Get ready for assignment 2 – Add additional pubkey

- Add additional public keys to an instance
 - *pubkey=\$(cat ~/mykeypair.pub)*
bastion=\$(openstack server show bastion -c addresses -f json | jq -r '.addresses["qh2-uom-internal"][]')
ssh -i ~/mykeypair ubuntu@\${bastion} "echo \${pubkey} >> ~/.ssh/authorized_keys"
- Too lazy to type the private key every time?
 - *ssh-add <path to your private key>*
 - *ssh ubuntu@\${bastion}*
 - For macOS only, add the key to keychain
 - *ssh-add -k <path to your private key>*



A Word About SSH Tunnelling

- SSH Tunnelling, aka. SSH Port Forwarding, is a method of transporting arbitrary networking data over an encrypted SSH connection.
 - *ssh -i <path-to-private-key> -L 8080:<remote>:80 ubuntu@<IP address>*
 - *ssh -i mykeypair -L 8080:127.0.0.1:80 ubuntu@192.168.0.2*





Something Good to Know – Bastion Host

- Bastion Host, aka. Jump host, is a host that is designed and positioned to withstand attacks. It is used to provide access to a private network from an external network.
 - `ssh -i <path-to-private-key> -J ubuntu@<bastion host> ubuntu@<target host>`
 - `ssh -i mykeypair -J ubuntu@128.250.55.1 ubuntu@192.168.0.2`

